

The allegory axioms, and what defines them in the Frobenius calculus

by

§1. Cartesian bicategory of relations

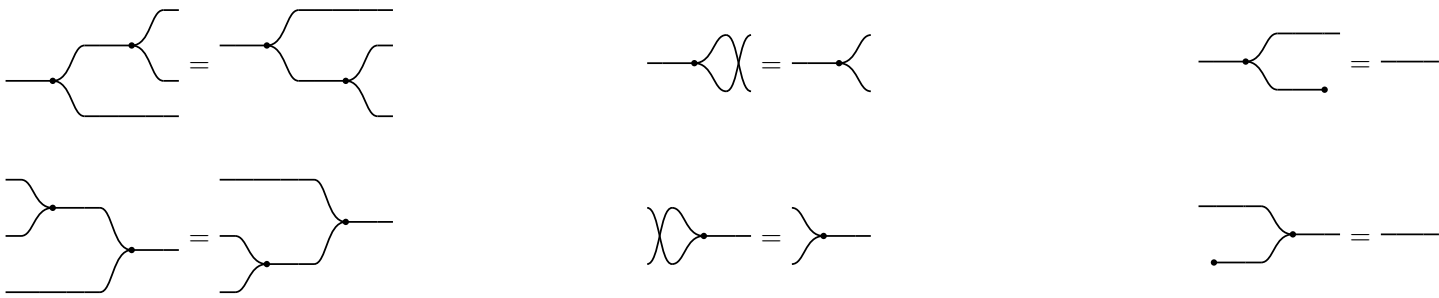
Definition 1.1.

A **cartesian bicategory of relations** is a poset-enriched category(bicategory) that is has a symmetric monoidal product $\otimes$ (cartisin product of set in Rel, vertical junctposition in the diagram) and

$\forall$ object A , $(A, \triangleleft, \multimap) \vdash (A, \triangleright, \multimap)$, **Frobenius**, arrows **lax**.

$(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow \mathbb{I})$ formed a cocommutative comonoid

$(\triangleright : A \otimes A \rightarrow A, \multimap : \mathbb{I} \rightarrow A)$ formed a commutative monoid



In $\text{Rel}(\text{Set})$, the four generators are these relations on a set A :

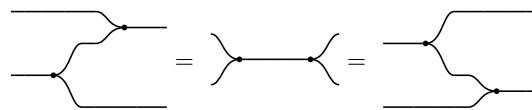
- $\triangleleft = \{(a, (a, a))\}$ — copy: one in, two identical out
- $\triangleright = \{((a, a), a)\}$ — merge: a pair passes only when its two components agree, and the common value comes out
- $\multimap = A \times \{*\}$ — discard: forget the value, keep only that there was one
- $\multimap = \{*\} \times A$ — create: any element at all

§1.1. $\forall$ object A , $(A, \triangleleft, \multimap) \vdash (A, \triangleright, \multimap)$



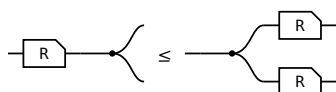
The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In Rel it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§1.2. The Frobenius law

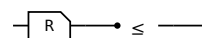


The only clause coupling the comonoid to the monoid beyond adjointness. Both sides reduce to the same picture, $\triangleright \triangleleft$ — merge, then copy — which is the two-in two-out **spider**. Every connected diagram built from these four generators collapses to the spider on its own inputs and outputs, and this equation is what starts that collapse. A **bubble** is the other order, $\triangleleft \triangleright$: copy then merge, a closed loop, which the idempotency of n uses below.

§1.3. Every arrow is a lax comonoid homomorphism



$$R \triangleleft \leq \triangleleft (R \otimes R)$$

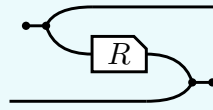


$$R \dashv \circ \leq \dashv \circ$$

§2. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

Definition 2.1.

The **converse** R° of $R : a \rightarrow b$ is R with both of its wires turned round,

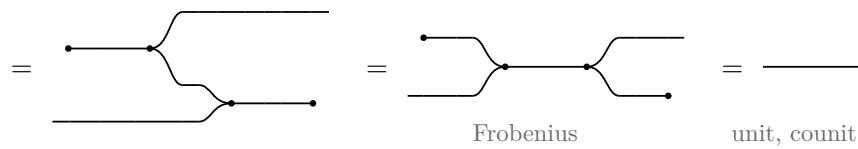


$$R^\circ = (\circlearrowleft \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \circlearrowright)$$

where $\circlearrowleft \triangleleft : \mathbb{I} \rightarrow a \otimes a$ opens a pair of wires out of nothing and $\triangleright \circlearrowright : b \otimes b \rightarrow \mathbb{I}$ closes one, so the input of R° is where the output of R was. And it is a contravariant 2-functor $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$:

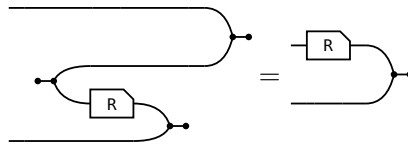
$$(i) \ 1^\circ = 1 \quad (ii) \ (R \ S)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

§2.1. $1^\circ = 1$ (snake)



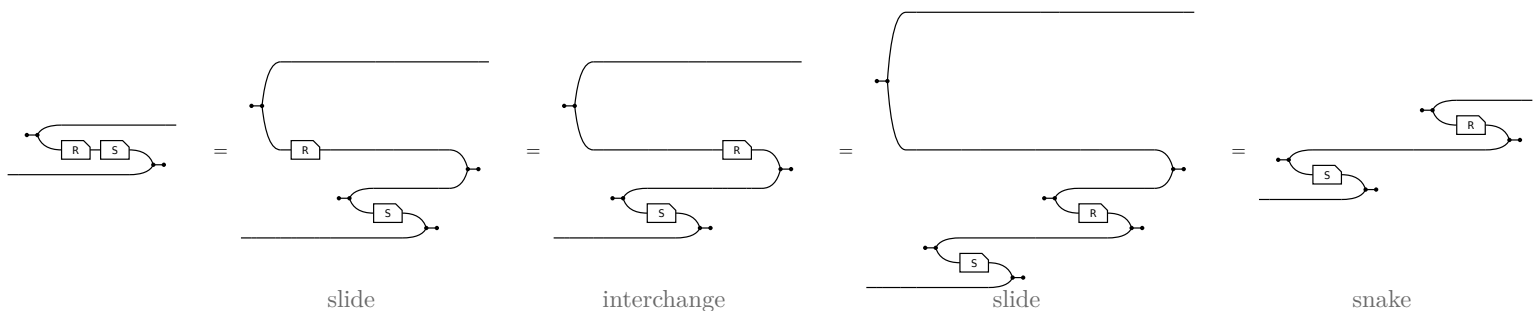
§2.2. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



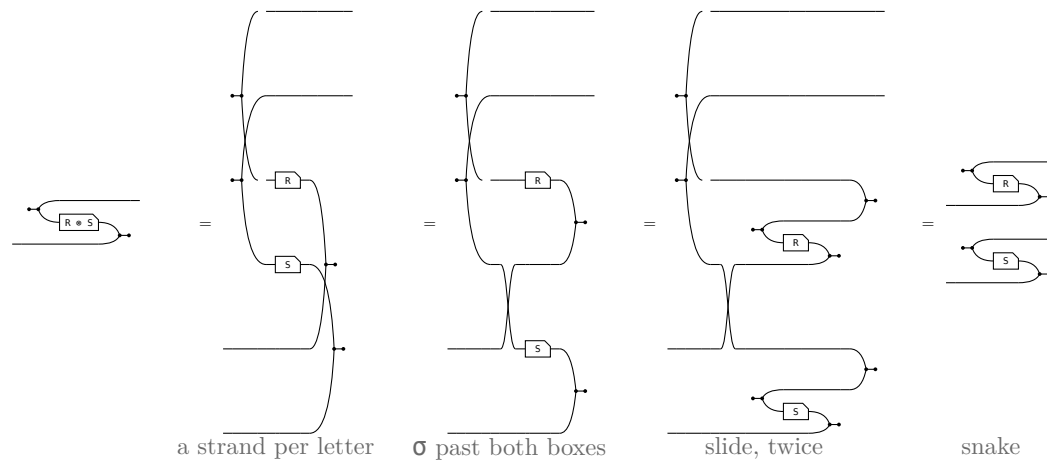
R° is DEFINED as the bending of $(R \otimes 1) \triangleright \circlearrowright$, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the $\circlearrowleft \triangleleft$ bends the a strand down and the $\triangleright \circlearrowright$ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

§2.3. $(R \ S)^\circ = S^\circ R^\circ$



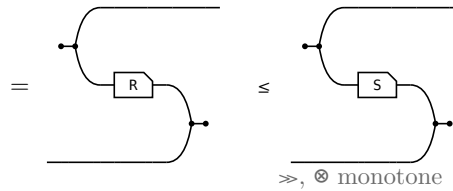
§2.4. $(R \otimes S)^\circ = R^\circ \otimes S^\circ$

A merge at a product is two merges behind a crossing, so the pair unbends one strand at a time and the crossing is all that is left to move.



§2.5. $R \leq S$ implies $R^\circ \leq S^\circ$

R sits in a frame of wires built from $\gg$ and $\otimes$, and both of those are monotone, so the box may be replaced where it stands.



§3. Every arrow is a lax monoid homomorphism

The merge/create mirror of the two comonoid laws opening this note — the paper's (41) and (42), p. 22, which it gets by bending them with (ii) and (iv) above. No bending is needed: expand by the unit $1 \leq \triangleleft \triangleright$, duplicate the box with the comonoid law, contract by the counit $\triangleright \triangleleft \leq 1$. Both halves of the adjunction, and nothing else.

In $\mathbf{Rel}(\mathbf{Set})$ with $R \subseteq A \times B$, where $\triangleright$ is the equality test on a pair and $\bullet$ produces any element at all:

$$\begin{aligned}
 \triangleright R &= \{((a, a), b) : a R b\} \text{ — two equal inputs, then one run of } R \\
 (R \otimes R) \triangleright &= \{((a, a'), b) : a R b \text{ and } a' R b\} \text{ — a run on each input, then the results merged} \\
 \bullet R &= \{(*, b) : \exists a. a R b\} \text{ — the image of } R \\
 \bullet &= \{*\} \times B \text{ — all of } B
 \end{aligned}$$

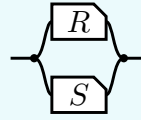
Take $a' = a$ for the first containment; every b lies in B for the second. Each is strict exactly when its reverse fails — R not injective, R not surjective — which is what the later table on maps reads off them.

inequation, and what proves it	picture
$(41) \triangleright R \leq (R \otimes R) \triangleright$ $\triangleright R = \triangleright R 1 \leq \triangleright R \triangleleft \triangleright \leq \triangleright \triangleleft (R \otimes R) \triangleright \leq (R \otimes R) \triangleright$ — unit, lax $\triangleleft$, counit.	
$(42) \bullet R \leq \bullet$ $\bullet R = \bullet R 1 \leq \bullet R \bullet \leq \bullet \bullet \leq \bullet$ — the same three steps at $\mathbb{I}$.	

§4. $\cap$ is a commutative idempotent monoid on every hom-set

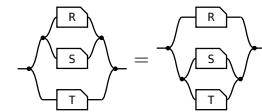
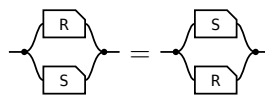
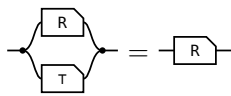
Definition 4.1.

The **meet** of $R, S : a \rightarrow b$, the paper's **convolution**, is $R \cap S := \triangleleft (R \otimes S) \triangleright$ — copy the input, run R and S on the two copies, merge the results — so what comes out is what both of them do.



transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow $\tau = \multimap \multimap$ (the paper's Lemma 4.11).

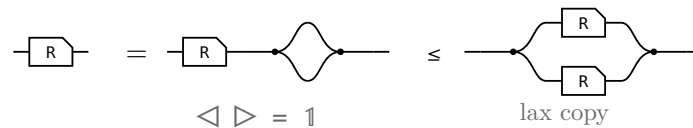


unit: one half of τ per end — the merge's unit law absorbs the $\multimap$, the copy's counit law the $\multimap$

commutative: σ crosses $R \otimes S$ by naturality and is absorbed by cocommutativity and commutativity

associative: coassociativity and associativity; $\otimes$ re-brackets for nothing, being strict here

§4.1. $R \cap R = R$, the one that is not bookkeeping



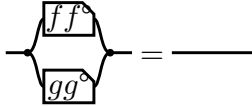
The last picture is $R \cap R$ itself, so this is $R \leq R \cap R$ and the lax copy law is the whole of it. The other direction is not proved here: $R \cap S \leq R$ holds for every S , by weakening S to τ and then collapsing $R \cap \tau$ — which is the paper's own remaining three steps, packaged once.

So $\leq$ is the order this monoid induces. $R \cap S \leq R$ comes from the unit, and idempotency turns anything under both S and T into something under $S \cap T$, since $R = R \cap R \leq S \cap T$.

§5. The eight axioms, and what proves each

axiom, and what supplies it	picture
$R (S \cap T) \sqsubseteq R S \cap R T$ — the lax copy law.	

§6. Unitary and pre-tabular — what the other direction needs

condition	allegory	Frobenius side
unitary	τ is a unit when 1τ is the largest endomorphism on it and every object carries an entire arrow to it. A unit is what makes τ exist.	free. 1 is the unit object, $1 \leq \multimap \multimap$ says $\multimap$ is entire, and $\tau = \multimap \multimap$ is maximal.
pre-tabular	f, g tabulate R when both are maps, $R = f \circ g$, and 	this is what $\otimes$ is. Given unitarity it reduces to tabulating each τ , and a tabulation of τ is an object $n \otimes m$ with its two projections. One side derives the product from tabulations, the other posits it.

condition	allegory	Frobenius side
	R is tabular when some pair tabulates it, the allegory pre-tabular when every arrow lies under a tabular one.	

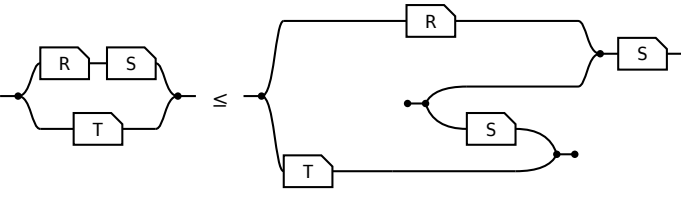
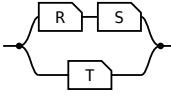
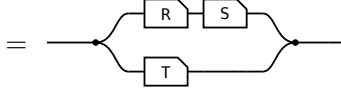
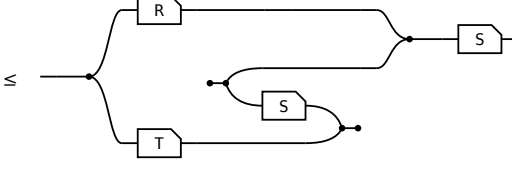
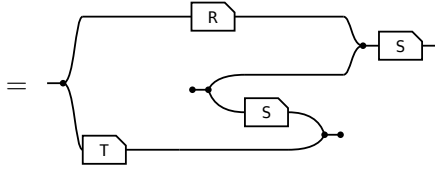
Together: **cartesian bicategory of relations** $\simeq$ **unitary pre-tabular allegory**.

§7. Union and residual

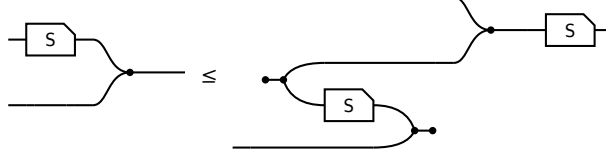
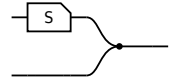
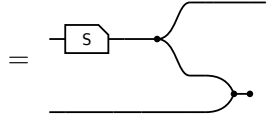
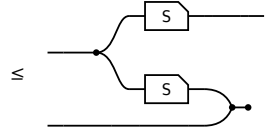
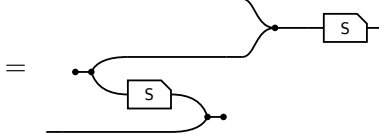
Both are past what the definition can build: a union needs a **biproduct** on top of the Frobenius structure, a residual a **second composition** with linear adjoints. A union draws as a **tape** — the rounded wrapper is the second product, and its fork and join open and close a branch a particle takes exactly one of. A residual draws as **long division**, not as its own term, which is a composition against a complement.

statement	picture
$R \sqsubseteq R \cup S$ union $R \ S := \langle R, S \rangle [1, 1]$: offer both branches, then forget which was taken.	
$R \cup S = S \cup R$	
$\perp \cup R = R$ $\perp$ routes through the zero object; there is no shape for it.	
$R (S \cup T) = R S \cup R T$ — composition distributes over union, which n does not.	
$(R \cup S)^\circ = R^\circ \cup S^\circ$	
$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$ — what makes the whole distributive , and the one picture with both layers in it.	
$(R / S) S \sqsubseteq R$ — the residual is a lower bound of the arrows it is the greatest of.	

§8. The modular law, from Frobenius

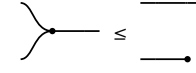
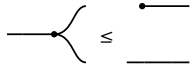
the modular law, $R \ S \cap T \leq (R \cap T \ S^\circ) \ S$ 		
term	picture	the rule that reaches it
$(R \ S) \cap T$		the start: $R \ S \cap T$.
$= (\triangleleft (R \otimes T)) ((S \otimes 1) \triangleright)$		Factor out the shared prefix: $(R \ S) \otimes T$ is $(R \otimes T) (S \otimes 1)$.
$\leq (\triangleleft (R \otimes T)) ((1 \otimes S^\circ) (\triangleright S))$		The merge slide. S slides through the merge and reappears as S° on the other strand. The only inequality in the proof.
$= (R \cap (T \ S^\circ)) \ S$		Fold the prefix back in; the copy-merge pair reads as a meet again.

S occurs once on the left of the slide and twice on the right, and the lax copy law is the only law in the definition that may duplicate a box — so that is where the inequality has to be.

the merge slide, $(S \otimes 1) \triangleright \leq (1 \otimes S^\circ) \triangleright S$ 		
term	picture	the rule that reaches it
$(S \otimes 1) \triangleright$		the start: $(S \otimes 1) \triangleright$.
$= ((S \triangleleft) \otimes 1) (1 \otimes \triangleright \bullet)$		Rebuild the merge from a copy and a right bracket, the shape the lax copy law applies to.
$\leq ((\triangleleft (S \otimes S)) \otimes 1) (1 \otimes \triangleright \bullet)$		The lax copy law: $S \triangleleft$ becomes $\triangleleft (S \otimes S)$, the box copied. The whole of the modular law, in one step.
$= (1 \otimes S^\circ) (\triangleright S)$		Reshape back, and the surviving duplicate slides round the right bracket to become S° .

§9. $R \leq R \circ R$, by cutting two wires

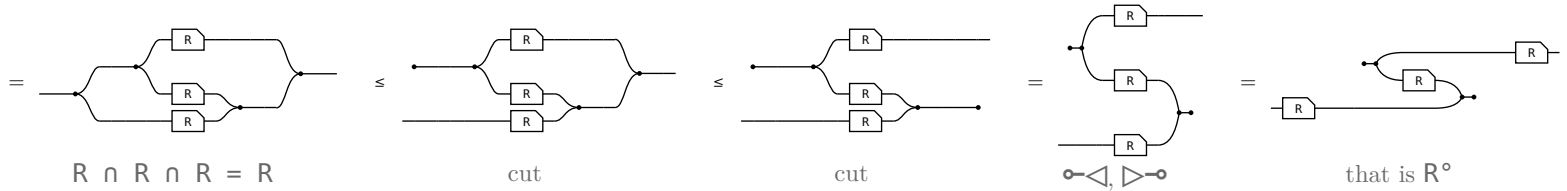
Freyd reads this off the modular law (§2.112): $R \sqsubseteq 1R \sqcap R \sqsubseteq (1 \sqcap R R^\circ) R \sqsubseteq R R^\circ R$. In pictures neither the modular law nor a meet with 1 is wanted. The one law that makes a picture bigger is spent twice, and it is the cut, $1 \leq \multimap \multimap$:



cut the copy's left output and what comes out there is **anything at all**

cut the merge's right input and it is **simply discarded**

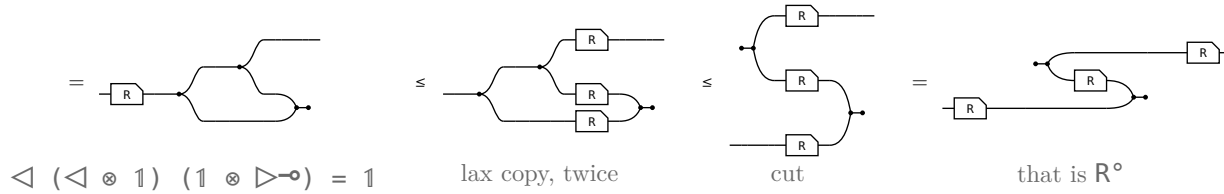
R is its own three-fold meet — three copies of the box between a copy tree and a merge tree — and the two cuts turn the trees into a $\multimap \sqcap$ and a $\sqcup \multimap$. The three boxes never move:



Read the last picture from the input, which is now the bottom strand: through the bottom box, up the $\sqcup \multimap$ and back along the middle box — which is that box turned round, so it is R° — down the $\multimap \sqcap$ and out along the top box. Where the meet had one input feeding all three boxes and one output draining all three, the zig-zag has both of those **cut**, and a cut wire is the one thing this calculus lets you add for free.

§10. The same, from the cap end

The section above cuts twice in its chain. This one cuts once and pays for the other cut with the **lax copy law** — $R \sqcap \leq \sqcap (R \otimes R)$, the only law in the definition that may duplicate a box. Read 1 as a copy tree whose last two legs are capped back:



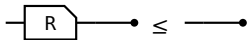
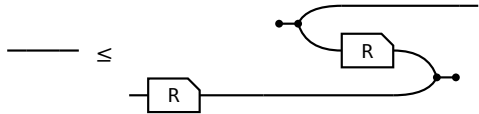
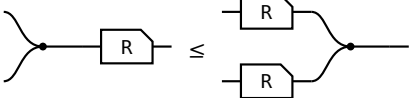
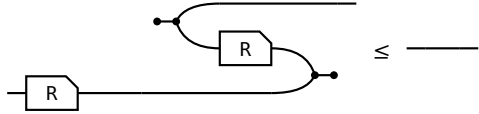
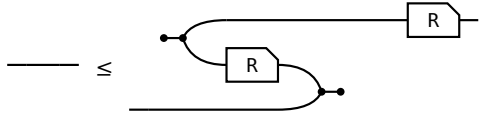
The three boxes are not there to begin with: there is one, and the two copy dots walk back **through** it, leaving it behind three times. So the $\sqcup \multimap$ is the frame's own cap before anything is cut, and the copy tree is the only thing left to cut.

Neither chain is cheaper in what it **spends**. $\sqcap \sqcup = 1$ is itself the merge cut, so the second $1 \leq \multimap \multimap$ is underneath this one; $R \sqcap R = R$ is the lax copy law, so that is underneath §9. Both routes pay both. What differs is the **layer**: this chain names only the copy, the cap and the box, so it holds as soon as the converse does — before $\sqcap$ exists. §9 climbs into the meet semilattice and back down to state a containment in which no meet occurs.

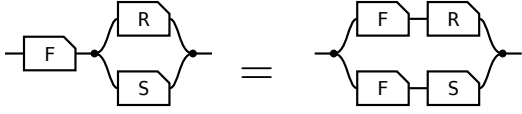
§11. Maps

Every arrow is lax for $\sqcap$ and for $\multimap$ by axiom, and for $\sqcup$ and $\multimap$ by the section above. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's **Simple** and **Entire**; that the two forms agree is a separate theorem, not proved here.

holds for every arrow	property	holds iff
$R \sqcap \leq \sqcap (R \otimes R)$ 	(SV) single valued $R^\circ R \sqsubseteq 1$	

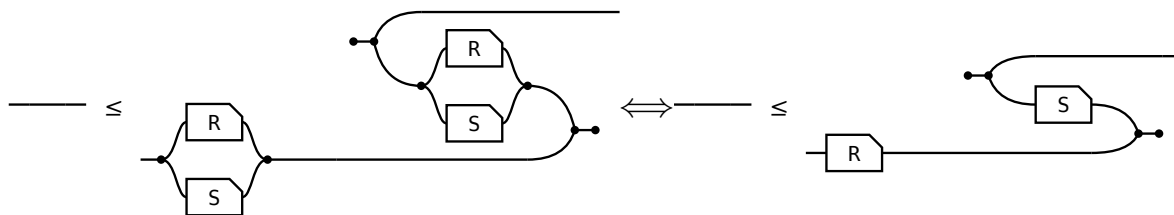
holds for every arrow	property	holds iff
$R \multimap \leq \multimap$ 	(TOT) total $1 \sqsubseteq R R^\circ$. With single valued, a map.	
$\triangleright R \leq (R \otimes R) \triangleright$ 	(INJ) injective $R R^\circ \sqsubseteq 1$	
$\multimap R \leq \multimap$ 	(SUR) surjective $1 \sqsubseteq R^\circ R$, that is, R° entire.	

What (SV) buys, in one law. Only single valuedness is spent: F may be partial, and entirety is never used.

law	picture
$F (R \cap S) = F R \cap F S$ F single valued F = shops Ann may go to, R = has milk, S = has bread. Single valued means one shop, so the sides agree.	 one shop with both milk at A, bread at B

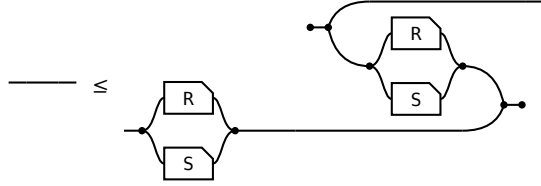
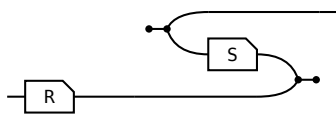
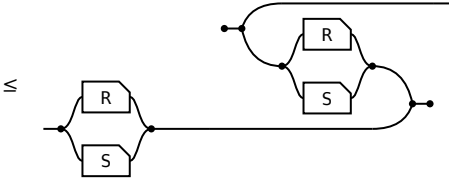
§12. Entireness of an intersection

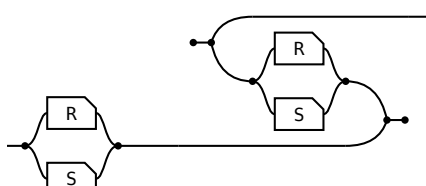
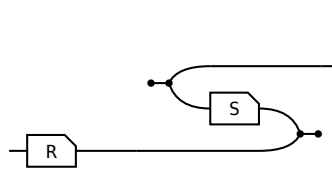
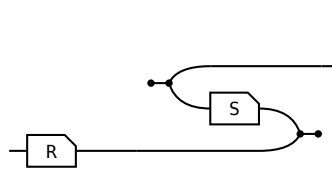
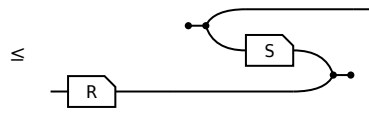
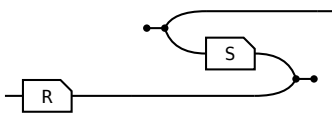
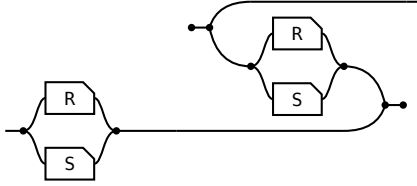
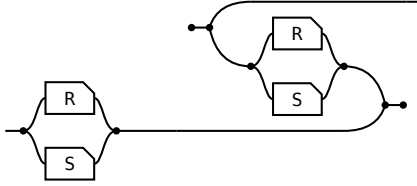
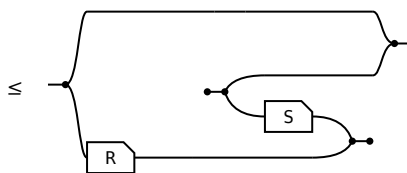
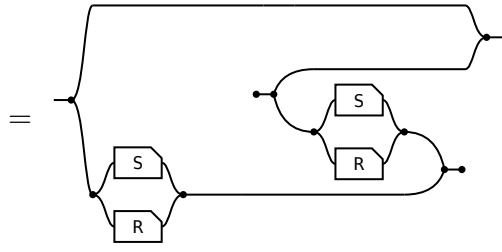
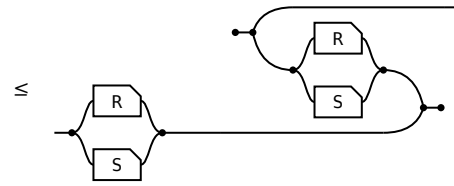
R is **entire** when $1 \sqsubseteq R R^\circ$, which is (TOT) above. When is an **intersection** entire?



$$\text{Total } (R \cap S) \leftrightarrow 1 \leq R S^\circ$$

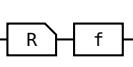
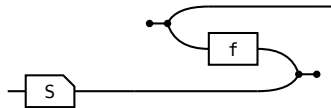
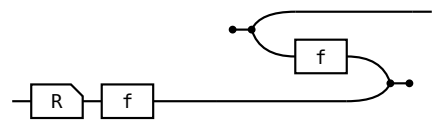
On the left $R \cap S$ is named **twice**; on the right R and S once each and the meet is gone. The meet is needed to **ask** whether something is entire, not to **answer** it. Where $\sqsubseteq$ is **defined** as $X \cap Y = X$ the two sides are one proposition; here $\leq$ is primitive, so both directions are real steps — and they cost very different things.

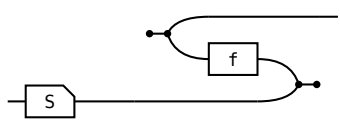
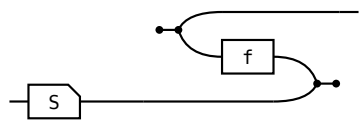
$\Rightarrow$ given $R \cap S$ entire,  show 		
term	picture	the rule that reaches it
1		the start: 1.
$\leq (R \cap S) (R \cap S)^\circ$		the hypothesis: $1 \leq P P^\circ$ at $P = R \cap S$.

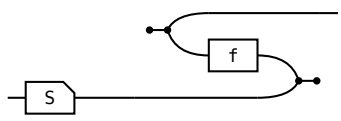
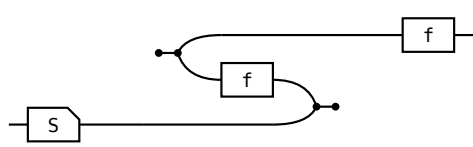
$\Rightarrow$ given $R \cap S$ entire,  $\leq$  show $\leq$ 		
term	picture	the rule that reaches it
$\leq R S^\circ$		monotonicity, twice. No modular law.
$\Leftarrow$ given  $\leq$ show $R \cap S$ entire,  $\leq$ 		
term	picture	the rule that reaches it
1		the start: 1.
$\leq 1 \cap (R S^\circ)$		the hypothesis, met with 1.
$= 1 \cap ((S \cap R) (S \cap R)^\circ)$		$1 \cap R S^\circ = 1 \cap (S \cap R) (S \cap R)^\circ$. $R S^\circ$ relates a to a' when some b has $a R b$ and $a' S b$; the $1 \cap$ sets $a' = a$, and it reads “some b has $a R b$ and $a S b$ ” — one arrow of $R \cap S$. Two arrows become one, and that is what needs the modular law.
$\leq (R \cap S) (R \cap S)^\circ$		$X \cap Y \leq Y$, then $S \cap R = R \cap S$.

§13. The shunting rule

A map moves from one side of a containment to the other at the cost of its converse, $R f \leq S \Leftrightarrow R \leq S f^\circ$. It is how a function gets out of the way so the relation underneath can be reasoned about. Each direction spends exactly one half of **map** — never both.

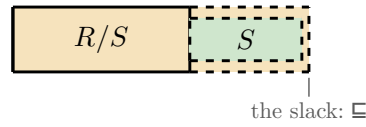
shunting right, $\Rightarrow$ given  $\leq$  show  $\leq$ 		
term	picture	the rule that reaches it
R		the start: R.
$\leq R (f f^\circ)$		total: $1 \leq f f^\circ$, so $f f^\circ$ may be inserted after R.

shunting right, $\Rightarrow$ given $\text{---} \boxed{R} \text{---} \boxed{f} \text{---} \leq \text{---} \boxed{S} \text{---}$ show $\text{---} \boxed{R} \text{---} \leq \text{---} \boxed{S} \text{---}$ 		
term	picture	the rule that reaches it
$\leq S f^\circ$	$\leq$ 	Re-bracket to $(R f) f^\circ$ and apply the hypothesis.

shunting right, $\Leftarrow$ given $\text{---} \boxed{R} \text{---} \leq \text{---} \boxed{S} \text{---}$ show $\text{---} \boxed{R} \text{---} \boxed{f} \text{---} \leq \text{---} \boxed{S} \text{---}$ 		
term	picture	the rule that reaches it
$R f$	$\text{---} \boxed{R} \text{---} \boxed{f} \text{---}$	the start: $R f$.
$\leq S (f^\circ f)$	$\leq$ 	Apply the hypothesis on the left; the two f s are now adjacent.
$\leq S$	$\leq \text{---} \boxed{S} \text{---}$	single valued: $f^\circ f \leq 1$, so the pair cancels.

Shunting left is the mirror, same shape, map on the other side throughout. What the rule is, is an adjunction: $f \dashv f^\circ$, with **total** the unit and **single valued** the counit, so “ f is a map” and “ f has a right adjoint, namely f° ” are one statement.

§14. Division

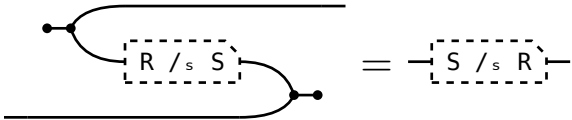


$T \sqsubseteq R/S \iff T S \sqsubseteq R$	
$T \sqsubseteq S \setminus R \iff S T \sqsubseteq R$	
$(R/S) S \sqsubseteq R$ Strict at $S = \emptyset$: R/S is everything, $(R/S) S = \emptyset$.	
$S (S \setminus R) \sqsubseteq R$	
associate: $R/(S_1 S_2) = (R/S_2)/S_1$	
$(S T) \setminus R = T \setminus (S \setminus R)$	
maps: $f (R/S) = (f R)/S$ The same three pieces bracketed two ways, which is the licence to write $f R / S$. The shunting rule.	
$R/(f S) = (R/S) f^\circ$ A map moves out of a denominator and reappears as f° outside the box. The shunting rule again.	
§2.314: $(R/S) (S/T) \sqsubseteq R/T$	
$1 \sqsubseteq R/R$ R = shop a stocks c , so R/R = “ a stocks all a ' does”, true of every shop. Strict: two shops stocking only milk contain each other and stay two shops.	
$(R/R) (R/R) = R/R$ Freyd writes $\sqsubseteq$, but with $1 \sqsubseteq R/R$ above it is an equality: $R/R = 1 (R/R) \sqsubseteq (R/R) (R/R)$ is the other direction. A preorder is idempotent.	
$(R/R) R = R$	
$R/1 = R$	
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$ A denominator's union becomes a numerator's meet: dividing by more leaves less.	
$S \setminus (R/T) = (S \setminus R)/T$ Which is why $S \setminus R/T$ needs no bracket.	

Fifteen laws, fifteen pictures, and not one shows a generator: $\cap$, $\cup$, $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

§15. Symmetric division

$R/_s S \triangleq (R/S) \cap (S/R)^\circ$, the two-sided quotient: everything R says about a column and everything S says about it, in both directions.

$(R/_s S)^\circ = S/_s R$ The meet is inside the definition, not the statement, so it stays under the label.	
$(R/_s S)(S/_s T) \sqsubseteq R/_s T$ $/_s$ is converse-symmetric and transitive, like an equality of columns.	